Two Programs, One Shared Record

Concurrency and the race condition

SCAN / LEARN

THE MAIN IDEA

When two operations touch the same data, their order can change the result. Synchronization protects shared state.

FOLLOW THE SYSTEM

01 START

A balance is 100.

02 PROGRAM A

Reads 100 and plans to add 10.

03 PROGRAM B

Also reads 100 and plans to subtract 20.

04 WRITE BACK

Without coordination, one update can erase the other.

READ 100 → TWO CHANGES → COLLISION → LOST UPDATE

COMMON MISCONCEPTION

Each operation is correct alone, so both together must be correct.

MORE ACCURATE

A transaction or other synchronization can keep updates consistent.

THREE QUESTIONS TO KEEP

PROBLEM: Who can change a shared record first?

FAILURE: A lost update gives the wrong balance.

IN REAL LIFE: Banking, inventory, reservations.

SOLO DISCOVERY

Try alone: start at 100, apply +10 and −20 in both orders; now imagine both read 100 first.